Contents

Volume 4 - Analytical Framework & Model Descriptions: Part G

Part G Road Map

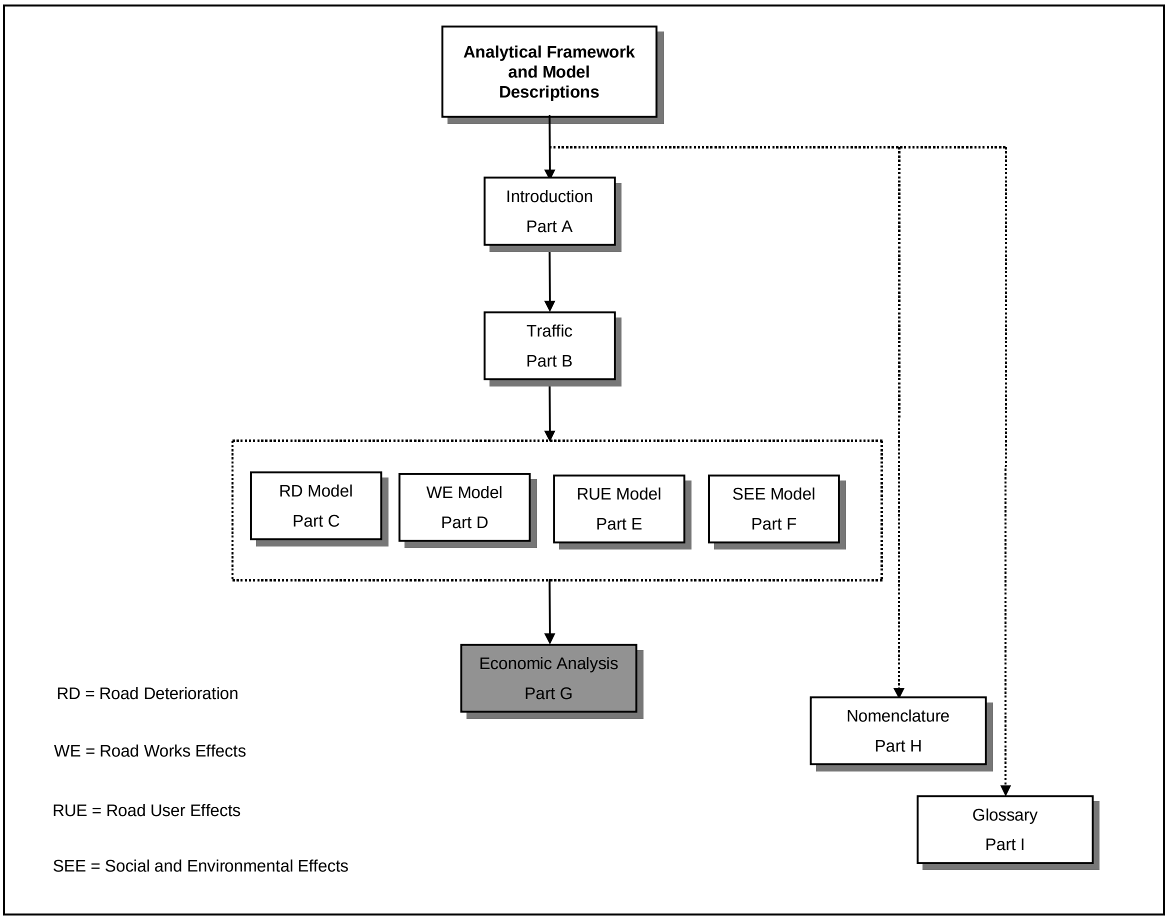

Figure G Analytical Framework and Model Descriptions Road Map

G1 Economic Analysis

1 Introduction

HDM-4 caters for three applications levels commonly used in decision making within the road sub-sector. The different applications, which are described in more detail in the Applications Guide, are:

1 **Strategic planning**

For estimating medium and long-term budget requirements for the development and preservation of a road network under various budgetary and economic scenarios.

2 **Programme analysis**

For preparing single or multi-year work programmes under budget constraints, in which those sections of the network likely to require maintenance, improvement, or new construction, are identified in a tactical planning exercise.

3 **Project analysis**

For estimating the economic or engineering viability of different road investment projects and associated environmental effects. Typical projects include the maintenance and rehabilitation of existing roads, widening or geometric improvement schemes, pavement upgrading and new road construction.

For all the three applications, the underlying operation of HDM-4 is based on the concept of life cycle analysis under a user-specified scenario of circumstances. This involves the analysis of pavement performance, road works effects and costs, together with estimates of road user costs and environmental effects, and economic comparisons of different project alternatives.

This chapter describes how HDM-4 is used to determine the benefits and costs associated with a road investment, and how these are applied in economic analysis and optimisation procedures to find the best use of available resources.

2 Background

2.1 Economic analysis

Economic analysis of the time stream of costs and benefits is used to compare the economic viability of different alternatives, and to provide the criteria needed for economic decision making. Decisions can be made about which option to implement, and when is the most favourable time for implementation. Economic analysis can also be used to investigate the technical standards and strategies to be followed by a particular investment decision. Economic analyses involves the following tasks:

1 Identification of the problem to be solved and the formulation of alternatives.

2 Identification and quantification of the life-cycle costs to be incurred and benefits to be realised.

3 Modelling future impacts of the proposed alternatives on road performance and traffic flow.

4 Economic comparison of the different alternatives, involving:

   (a) discounting the annual costs and benefits streams to a chosen base year

   (b) Comparing the time stream of costs for each pair of alternatives

   (c) Calculating the economic indicators such as the net present value, internal rate of return, benefit-cost ratio, and first year benefits

A project analysis usually involves a small number of road links or sections and the results of economic analysis would provide adequate information for decision making, since a budget would normally already have been approved for these activities.

2.2 Optimisation

The purpose of the Strategy and Programme applications is to calculate the economic benefits derived from maintenance or improvement options, and to select the set of investments to be made on a number of road sections within a network which will optimise an objective function.

Programme analysis is concerned with short to medium term planning and preparation where budget levels are known with reasonable certainty and the objective is to select a set of road sections and road works within the budget constraint.

Strategy analysis involves the analysis of an entire road network (or sub-network). The objective is either to determine which types of road works should be applied in order to maximise economic benefits, or it may also be applied to determine the budget required for a given long term target road network condition. Thus, the problem can be posed as one of searching for the combination of investment alternatives that optimises the objective function under a budget constraint or a road network condition constraint. Note that the set of investment options to be optimised is user-defined and is not the set of all possible options for the particular network; hence the problem is not true optimisation since all possible solutions are not normally considered. Note also that the investment options on any one road section are mutually exclusive.

The three alternative objective functions provided for the Strategy and Programme applications are:

1 **Maximisation of economic benefits** (that is, NPV)

This option is used when the problem can be defined as the selection of a combination of investment options applied on several road sections which maximises the NPV (net present value) for the whole network subject to the sum of the financial costs being less than the budget available.

2 **Maximisation of the improvement in network condition**

The roughness reduction on each road section multiplied by the section length (ΔIRI*Length) is used instead of NPV. Consequently, the arithmetic procedure is similar to that used for maximisation of economic benefits.

3 **Minimise costs of road works to achieve a given target road network condition**

This option is used mainly in the Strategy analysis application. The target road condition defined in terms of the long-term average roughness (IRI) over the whole analysis period must be specified for each road section. The optimisation procedure is then reduced to a simple selection of the road work options for which the average IRI (over the analysis period) is equal to or just below the target IRI and has the lowest total financial cost.

2.3 Classification of benefits and costs

Costs and benefits due to road investments may be classified into the following three broad categories:

1 **Benefits and costs expressed in monetary terms**

For example, vehicle operating costs, savings in travel time, accident costs.

2 **Quantified benefits and costs not expressed in monetary terms**

For example, road safety, pollution from vehicle emissions and traffic noise.

3 **Non-quantified benefits and costs**

For example, better social welfare, ecological impacts.

An economic analysis considers directly only benefits and costs expressed in monetary terms. Other costs and benefits may also need to be considered, and this is sometimes done within the framework of a multi-criteria analysis.

3 Benefits and costs considered in HDM-4

3.1 Summary of benefits and costs

HDM-4 considers quantified benefits and costs that can be expressed in monetary terms, and has some scope for considering those that cannot be expressed in this way. The benefits and costs considered are:

- **Costs incurred by the road administration** (see Section 3.2)
- **Road user costs** (see Section 3.3)
- **Environmental effects** (see Section 3.4)
- **Other benefits and costs** (see Section 3.5)

3.2 Costs incurred by the road administration

These costs are also referred to as road agency costs and include the following:

- **Road development**
- **Pavement maintenance**
- **Road-side or off-carriageway activities**

The cost of works is derived from the product of the physical quantities involved in the activity and the unit cost. These are determined for each road section and investment option, and for each year of the analysis period. The resulting costs are assigned to budget categories that are user-definable. The following default categories are used in HDM-4:

- **Capital (or periodic)**
- **Recurrent (or routine)**
- **Special**

Budget constraints can be applied separately to each category when required by the economic analysis and optimisation.

3.3 Road user costs

The following components of Road User Costs are modelled:

- **Motorised vehicle operating costs**

  These costs include:

  - Fuel and lubricant consumption
  - Tyre and parts consumption
  - Labour
  - Capital
  - Crew

- Overheads

- **Travel time costs**

  These costs include passenger travel time costs, and cargo holding time costs.

- **Non-motorised transport (NMT)**

  These costs include time and operating costs.

- **Accident costs**

  These costs are evaluated both in monetary and non-monetary terms, and separated into several different types (for example, fatal, injury, and damage only, and all accidents). Note users are allowed the flexibility to include or exclude accident costs from an economic analysis.

3.4 Environmental effects

The following environmental impacts are determined:

- **Vehicle emissions**
- **Energy use**
- **Traffic noise** (not included in this release)

3.5 Other benefits and costs

The user can specify those benefits and costs that are not modelled for each year of the analysis period. These benefits and costs are discounted and added to those that are calculated internally. These other benefits and costs are sometimes termed **exogenous**.

3.6 Unit costs

Unit costs are applied to the calculated physical and operational quantities to produce the cost estimates used in investment decisions and budget preparation. Unit costs should be expressed in economic terms when economic analysis is being undertaken, and in financial terms for financial analysis. Financial unit costs are the market prices of resources. Economic unit costs are the real value or opportunity costs of resources, and they are found by removing distortions such as taxes, subsidies and other miscellaneous costs from the market prices.

Unit costs are required for the following:

- **Road development, maintenance and road-side activities**

  These unit costs are specified by the user (see Part D).

- **Road user costs**

  These unit costs include vehicle resources, travel-time values, and road accident resources (see Part E).

In most cases, unit costs are specified in units-per-quantity. However, some costs are specified as a proportion of other costs, or as a lump sum.

In addition to calculating economic costs, financial costs are also computed if the user gives appropriate inputs (for example, unit costs in financial terms).

4 Outline methodology

4.1 Basic unit of analysis

The basic unit of analysis in HDM-4 is the **homogeneous** road section. Several investment options can be assigned to a road section for analysis. One or more vehicle types that use the road must also be defined together with the traffic volume specified in terms of the annual average daily traffic (AADT).

4.2 Life cycle analysis

The underlying operation of HDM-4 is common for the project, programme or strategy applications. In each case, HDM-4 predicts the life cycle pavement performance and the resulting user costs under specified maintenance and/or road improvement scenarios. The broad concept of the life cycle analysis is illustrated in Figure G1.1. The agency and user costs are determined by first predicting physical quantities of resource consumption and then multiplying these by the corresponding unit costs.

Two or more options comprising different road maintenance and/or improvement works should be specified for each candidate road section with one option designated as the **do minimum** or **base case** (usually representing minimal routine maintenance). The benefits derived from implementation of other options are calculated over a specified analysis period by comparing the predicted economic cost streams in each year against that for the respective year of the base case option. The discounted total economic cost difference is defined as the net present value (NPV). The average life cycle riding quality measured in terms of the international roughness index (IRI) is also calculated for each option.

INPUTS	MODEL	OUTPUTS
Vehicle type, volume, growth, loading, physical parameters, terrain, precipitation, road geometry, pavement characteristics, unit costs	Start of analysis loop	
Pavement type, strength, age, condition, and ESAL	Road Deterioration	Cracking, ravelling, pot-holes, rut depth, faulting (paved); gravel thickness (unpaved); roughness
Road geometry and roughness; vehicle speed, type; congestion parameters; unit costs	Road User Effects	Fuel, lubricant, tyres, maintenance, fixed costs, speed, travel time, road user costs
Road works standards and strategies	Works Effects	Reset cracking, ravelling, pot-holes, rut depth (paved); gravel thickness (unpaved); roughness, works quantities and agency costs
Road geometry and surface texture, vehicle characteristics	Social and Environmental Effects	Levels of emissions and energy used, and number of accidents
Developmental, accident, environmental, and other exogenous costs and benefits	Economic Analysis	Costs and benefits, including exogenous benefits
	Return to start of analysis loop	Total costs by component; net present values and rates of return by section

Figure G1.1 Life cycle analysis procedure in HDM-4

4.3 Models

Total life-cycle conditions and costs of sections or road networks can be simulated over a user-defined period into the future. The inter-dependence between the costs incurred by the road administration and the road user is recognised, and models are used to predict cost streams under the various headings.

The models incorporated in HDM-4 contain technical relationships for the following purposes:

1 Calculation of traffic volumes and flows, and vehicle loading over the road section.

2 Prediction of road deterioration, and works effects and costs, that are incurred in response to traffic flows, time and the surrounding environment.

3 Prediction of the costs of road use incurred as road condition and traffic flow change over time.

4 Prediction of accident rates as a function of the road and traffic characteristics, and the evaluation of accident costs.

5 Evaluation of vehicle emissions and energy use due to different road investment projects.

6 Economic analysis by comparison of the impacts or effects of different road investment project alternatives.

4.4 Analysis sequence

The overall logic sequence for economic analysis and optimisation is illustrated in Figure G1.2a and represented below by pseudo codes. This shows the following:

1 **The outer analysis loop**

Enabling economic comparisons to be made for each pair of investment options, using the effects and costs calculated over the analysis period for each option, and indicates that generated and diverted traffic levels may vary depending on the investment option considered.

2 **Costs**

How annual costs to the road administration and to the road users are calculated for individual road section options.

3 **Optimisation procedures**

These are performed after economic benefits of all the section options have been determined.

The pseudo code that represents the outer analysis loop is given below:

```
START
    Define input data
    Loop for each option
        Loop for each section
            Loop for each analysis year
                Calculate traffic over the road section
                Model annual effects and costs (see Figure G1.2b)
                Store results for evaluation and reporting phase
            End loop
        End loop
    End loop
    Loop for each pair of options to be compared
        Loop for each analysis year
            Loop for each section
                Calculate non-discounted net benefits
                Calculate discounted net benefits
                Calculate net environmental effects and energy used
            End loop
            Calculate total non-discounted net benefits over all the sections (see Figure
            G1.2c)
            Calculate total discounted net benefits over all the sections (see Figure G1.2c)
            Calculate total environmental effects and energy used over all the sections (see
            Figure G1.2c)
        End loop
        Calculate economic indicators (NPV, IRR, BCR, and FYB see Section 5.3.1 and
        Figure G1.2c)
    End Loop
    Perform budget optimisation (for strategy and programme analysis)
    Output results
END
```

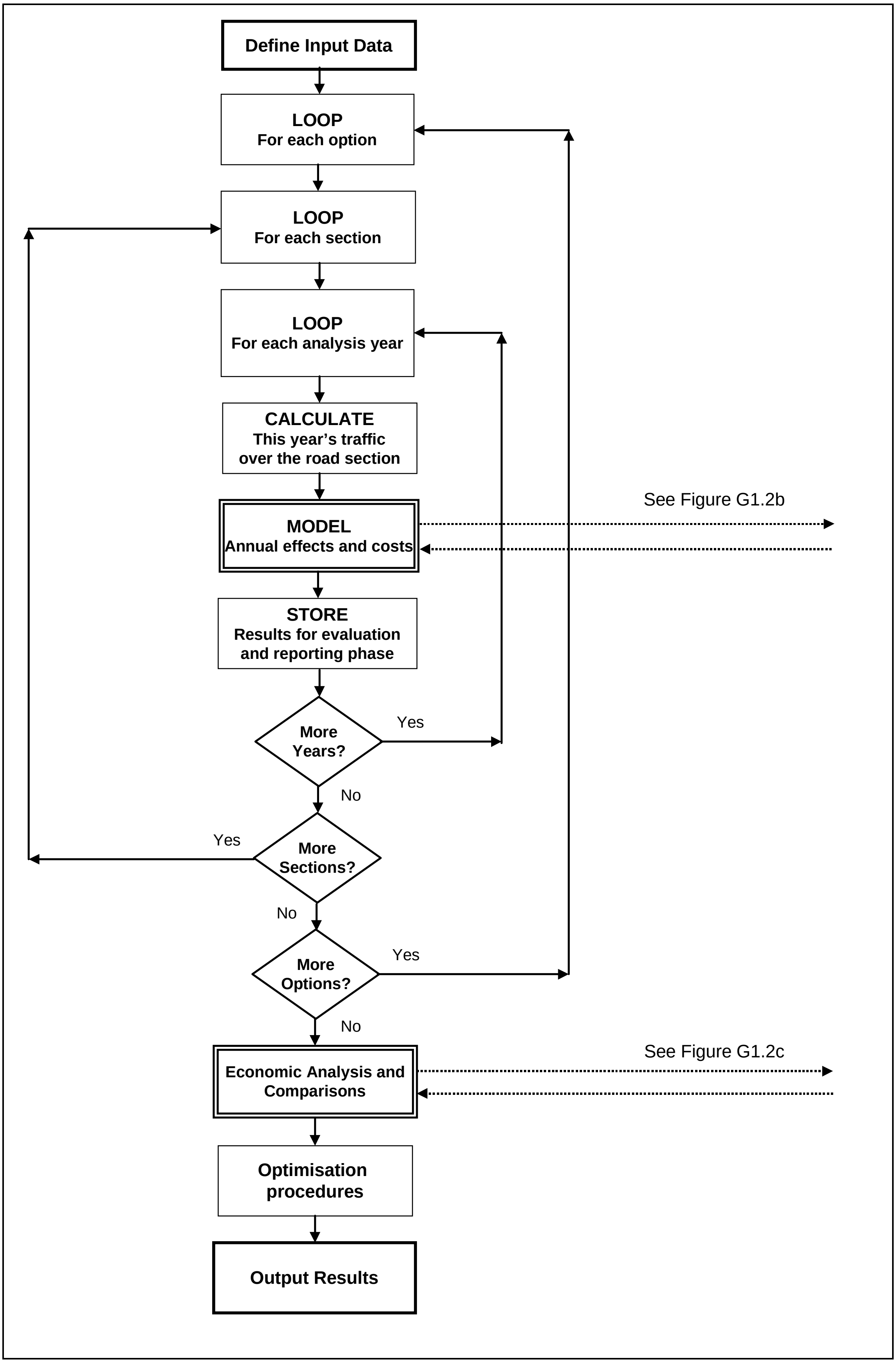

Figure G1.2a Overall analysis sequence logic – Part A

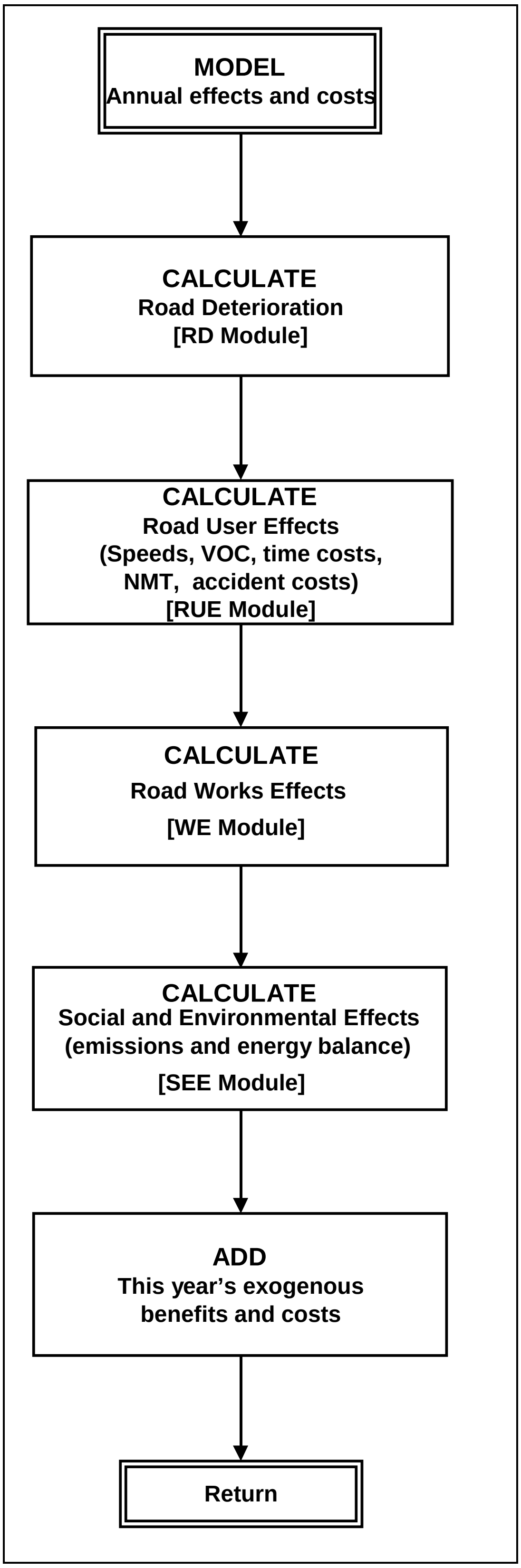

Figure G1.2b Overall Analysis sequence logic - Part B

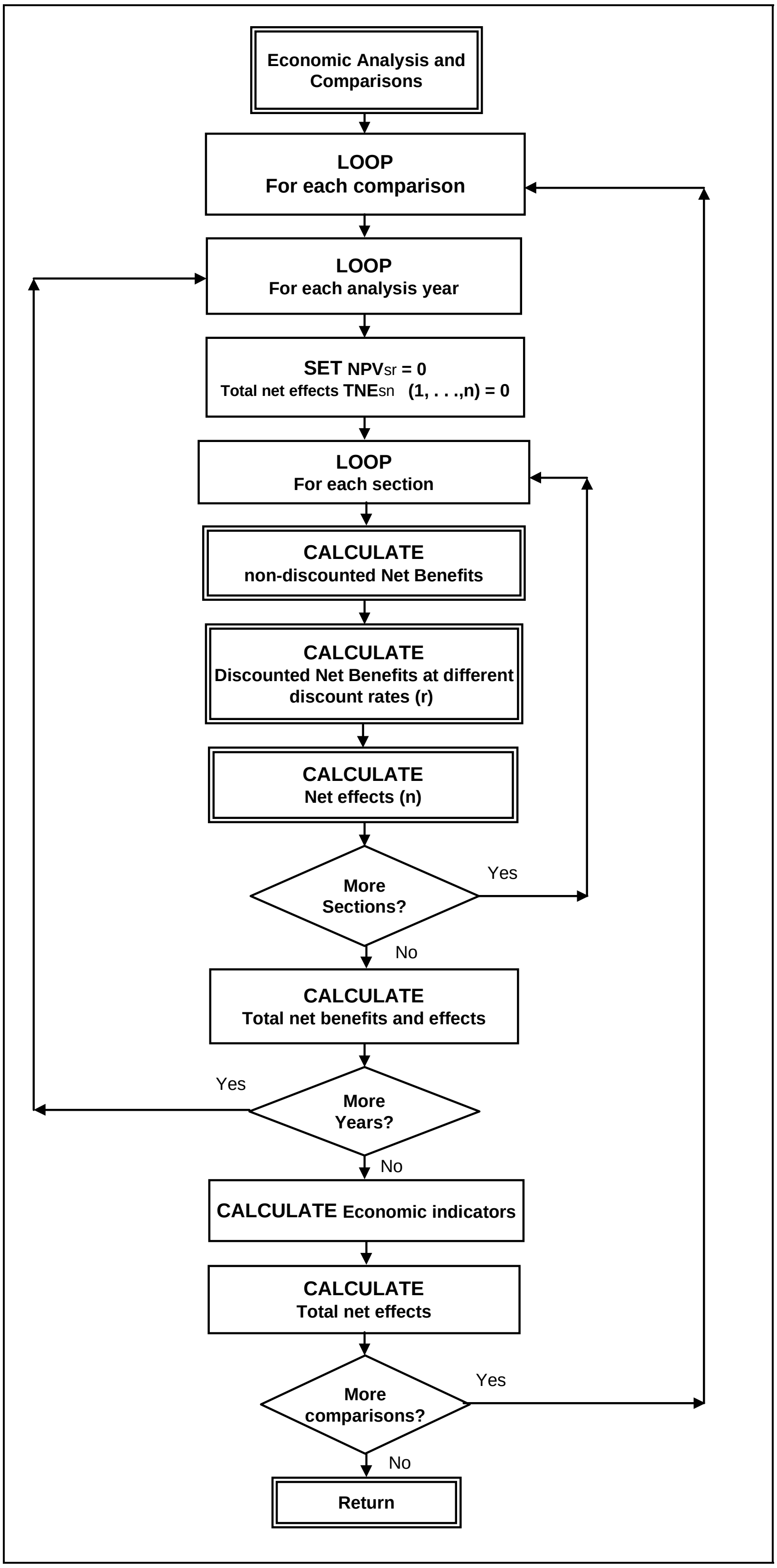

Figure G1.2c Overall Analysis sequence logic - Part C

The procedure for calculating annual road agency costs and road user effects for individual section options is illustrated in Figure G1.2b and summarised by the following steps:

1 **Calculate road deterioration** - in the RD module (see Part C)

2 **Calculate road user costs**

VOC, travel time costs, NMT time and operating costs, and accident costs - in the RUE module (see Part E).

3 **Calculate quantities and costs for road works** - in the WE module (see Part D)

4 **Calculate environmental effects**

For example, emissions and energy use - in the SEE module (see Part F).

5 **Add exogenous benefits and costs**

Figure G1.2c illustrates the inner analysis loops for economic analysis and comparison of each pair of road alternatives.

5 Economic analysis

5.1 Comparison of investment options

Economic indicators are computed at different user-specified discount rates using the time streams of benefits or costs resulting from the various comparative pairs of investment options. The term **investment options** has been used in this document to refer to both project options (or alternatives) and section options (or alternatives).

For each pair of investment options to be compared, the net benefits (or costs) of implementing one option relative to the other is calculated year by year. The various methods of comparison are described in sub-sections within Sections 5.2 and 5.3. In all cases, investment option *m* is compared against option *n* (that is, option *n* is the base case).

5.2 Determination of costs and benefits

5.2.1 Costs to the road administration

The cost differences between a pair of investment options, *m* and *n*, in a given year, are calculated as follows:

$$\Delta C_{(m-n)i} = \left[\sum_s C_{mis} - \sum_s C_{nis}\right] \qquad \text{...(5.1)}$$

where:

$\Delta C_{(m-n)i}$ the difference in road administration cost of investment option *m* relative to base option *n* for budget category *i*

C_{jis} the total costs to road administration incurred by investment option *j* (where *j* = *n* or *m*) for budget category *i*, for road section *s* (see Part D)

The difference in annual costs to road administration is given by the expression:

$$\Delta RAC_{(m-n)} = \sum_i \Delta C_{(m-n)i} \qquad \text{...(5.2)}$$

where:

$\Delta RAC_{(m-n)}$ the difference in annual costs to road administration of investment option *m* relative to base option *n*. (The summation is over all the budget categories)

These cost differences provide a relative measure of the increase in costs to the road administration, of implementing investment option *m* over base option *n*.

The difference in the salvage values of works performed under investment options *m* and *n* is a component of the net economic benefits to be included in the last year of the analysis period (see Section 5.2.4), and is given as:

$$\Delta SALVA_{(m-n)} = [SALVA_m - SALVA_n] \qquad \text{...(5.3)}$$

where:

$\Delta SALVA_{(m-n)}$	the difference in salvage value of implementing investment option *m* relative to base option *n*
$SALVA_j$	salvage value of the works performed under investment option *j* (where *j* = *n* or *m*) (see Part D)

5.2.2 Savings in road user costs

The annual economic benefits in terms of savings in road user costs are calculated separately by components and traffic categories as follows:

- **Savings in motorised vehicle operating costs**

Vehicle operating benefits due to normal and diverted traffic is calculated as follows:

$$\Delta VCN_{(m-n)} = \left[\sum_{s} VCN_{ns} - \sum_{s} VCN_{ms} \right] \quad ...(5.4)$$

$$VCN_{ns} = \sum_{k} TN_{nsk} * UC_{nsk} \quad ...(5.5)$$

$$VCN_{ms} = \sum_{k} TN_{msk} * UC_{msk} \quad ...(5.6)$$

Vehicle operating benefits due to generated traffic is calculated as follows:

$$\Delta VCG_{(m-n)} = \left[\sum_{s} \sum_{k} \{0.5 * [TG_{msk} + TG_{nsk}] * [UC_{nsk} - UC_{msk}]\} \right] \quad ...(5.7)$$

The summations are over all the motorised vehicle types (*k* = 1, 2, ..., K) specified by the user, and all road sections (*s* = 1, 2, ..., S) being analysed.

The annual saving in vehicle operating costs is given by the expression:

$$\Delta VOC_{(m-n)} = [\ \Delta VCN_{(m-n)} + \Delta VCG_{(m-n)}\ ] \quad ...(5.8)$$

where:

$\Delta VCN_{(m-n)}$	vehicle operating benefits due to normal and diverted traffic of investment option *m* relative to base option *n*
VCN_{js}	annual vehicle operating cost due to normal and diverted traffic over the road section *s* with investment option *j*
TN_{jsk}	normal and diverted traffic, in number of vehicles per year in both directions on road *s*, investment option *j*, for vehicle type *k*
UC_{jsk}	annual average operating cost per vehicle-trip over road section *s*, for vehicle type *k* under investment option *j* (where *j* = *n* or *m*)
VCG_{js}	annual vehicle operating cost due to generated traffic over road section *s* with investment option *j*

$\Delta VCG_{(m-n)}$	vehicle operating benefits due to generated traffic of investment option *m* relative to base option *n*
TG_{jsk}	generated traffic, in number of vehicles per year in both directions on road *s*, for vehicle type *k*, due to investment option *j*
$\Delta VOC_{(m-n)}$	savings in vehicle operating costs due to the total traffic of investment option *m* relative to base option *n*

- **Savings in travel time costs – motorised vehicles**

Vehicle travel time benefits due to normal and diverted traffic are calculated as follows:

$$\Delta TCN_{(m-n)} = \left[\sum_{s} TCN_{ns} - \sum_{s} TCN_{ms}\right] \quad ...(5.9)$$

$$TCN_{ns} = \sum_{k} TN_{nsk} * UT_{nsk} \quad ...(5.10)$$

$$TCN_{ms} = \sum_{k} TN_{msk} * UT_{msk} \quad ...(5.11)$$

Vehicle travel time benefits due to generated traffic are calculated as follows:

$$\Delta TCG_{(m-n)} = \left[\sum_{s}\sum_{k}\{0.5 * [TG_{msk} + TG_{nsk}] * [UT_{nsk} - UT_{msk}]\}\right] \quad ...(5.12)$$

The annual savings in travel time costs are given by the expression:

$$\Delta TTC_{(m-n)} = [\Delta TCN_{(m-n)} + \Delta TCG_{(m-n)}] \quad ...(5.13)$$

where:

$\Delta TCN_{(m-n)}$	travel time benefits due to normal and diverted traffic of investment option *m* relative to base option *n*
TCN_{js}	annual vehicle travel time cost due to normal and diverted traffic over road section *s* with investment option *j*
UT_{jsk}	annual average travel time cost per vehicle-trip over the road section *s*, for vehicle type *k*, under investment option *j* (where *j* = *n* or *m*)
TCG_{js}	annual vehicle travel time cost due to generated traffic over road section *s* with investment option *j*
$\Delta TCG_{(m-n)}$	travel time benefits due to generated traffic of investment option *m* relative to base option *n* on the given road section in the given year
$\Delta TTC_{(m-n)}$	savings in travel time costs due to total traffic of investment option *m* relative to base option *n*

- **Savings in NMT time and operating costs**

Non-Motorised Transport (NMT) time and operating benefits due to normal and diverted traffic are calculated as follows:

$$\Delta TOCN_{(m-n)} = \left[\sum_s TOCN_{ns} - \sum_s TOCN_{ms}\right] \quad ...(5.14)$$

$$TOCN_{ns} = \sum_k TN_{nsk} * UTOC_{nsk} \quad ...(5.15)$$

$$TOCN_{ms} = \sum_k TN_{msk} * UTOC_{msk} \quad ...(5.16)$$

NMT time and operating benefits due to generated traffic are calculated as follows:

$$\Delta TOCG_{(m-n)} = \left\{\sum_s \sum_k [0.5 * (TG_{msk} + TG_{nsk}) * (UTOC_{nsk} - UTOC_{msk})]\right\}$$

...(5.17)

The summations are over all the NMT types (k = 1, 2, ..., K) specified by the user, and all road sections (s = 1, 2, ..., S) being analysed.

The annual savings in NMT time and operating costs are given by the expression:

$$\Delta NMTOC_{(m-n)} = [\Delta TOCN_{(m-n)} + \Delta TOCG_{(m-n)}] \quad ...(5.18)$$

where:

$\Delta TOCN_{(m-n)}$	NMT time and operating benefits due to normal and diverted traffic of investment option m relative to base option n
$TOCN_{js}$	annual NMT time and operating costs due to normal and diverted traffic over the road section s with investment option j
TN_{jsk}	NMT normal and diverted traffic, in number of vehicles per year in both directions on road s investment option j, for vehicle type k
$UTOC_{jsk}$	annual average NMT time and operating cost per vehicle-trip over road section s, for vehicle type k, under investment option j (where j = n or m)
$TOCG_{js}$	annual NMT time and operating costs due to generated traffic over road section s with investment option j
TG_{jsk}	NMT generated traffic, in number of vehicles per year in both directions on road s, for vehicle type k, due to investment option j
$\Delta TOCG_{(m-n)}$	NMT time and operating benefits due to generated traffic of investment option m relative to base option n
$\Delta NMTOC_{(m-n)}$	annual savings in NMT time and operating costs due to total traffic of investment option m relative to base option n

- **Reduction in accident costs**

The benefits from reduction in total accident costs are given by the expression:

$$\Delta ACC_{(m-n)} = [AC_n - AC_m] \quad ...(5.19)$$

where:

$\Delta ACC_{(m\text{-}n)}$	the accident reduction benefits due to implementing investment option *m* relative to base option *n*
AC_j	the total accident costs under investment option *j* (where *j* = *n* or *m*)

- **Road user benefits**

The annual savings in road user costs are given by the expression:

$$\Delta RUC_{(m-n)} = \left[\Delta VOC_{(m\text{-}n)} + \Delta TTC_{(m\text{-}n)} + \Delta NMTOC_{(m\text{-}n)} + \Delta ACC_{(m\text{-}n)}\right] \qquad ...(5.20)$$

where:

$\Delta RUC_{(m\text{-}n)}$	the total road user benefits of investment option *m* relative to base option *n*

5.2.3 Other benefits and costs

The difference in other (exogenous) benefits and costs, for each pair of investment options *m* and *n* in a given year, are calculated as follows:

$$\Delta NEXB_{y(m-n)} = \left[EXB_{ym} - EXC_{ym} - EXB_{yn} + EXC_{yn}\right] \qquad ...(5.21)$$

where:

$\Delta NEXB_{y(m\text{-}n)}$	the annual net exogenous benefits of investment option *m* relative to base option *n*, in year *y*
EXB_{jy}	exogenous benefits for investment option *j*, in year *y*, (where *j* = *n* or *m*)
EXC_{jy}	exogenous costs for investment option *j*, in year *y*

5.2.4 Annual net economic benefits

For each pair of investment options, the annual net economic benefits of implementing option *m* relative to option *n* is obtained by combining the differences in costs to the road administration, road user costs, and other benefits and costs, as follows:

$$NB_{y(m\text{-}n)} = \left[\Delta RUC_{y(m\text{-}n)} + \Delta NEXB_{y(m\text{-}n)} - \Delta RAC_{y(m\text{-}n)}\right] \qquad ...(5.22)$$

where:

$NB_{y(m\text{-}n)}$	net economic benefit of investment option *m* relative to base option *n* in year *y*, and the parameters on the right-hand side are as defined earlier, but with subscript *y* added to indicate year

In the last year of the analysis period, the net economic benefits of implementing option *m* relative to option *n* is calculated as:

$$NB_{Y(m\text{-}n)} = \left[\Delta RUC_{Y(m\text{-}n)} + \Delta NEXB_{Y(m\text{-}n)} - \Delta RAC_{Y(m\text{-}n)} + \Delta SALVA_{(m\text{-}n)}\right] \qquad \text{...(5.23)}$$

where:

$NB_{Y(m\text{-}n)}$	net economic benefit of investment option *m* relative to base option *n* in the last year of the analysis period *Y*, and the parameters on the right-hand side are as defined earlier, but with subscript *Y* added to indicate the last year of the analysis period

5.2.5 New road sections (or links)

For the analysis of a new road section, the following variables used in the equations given in Sections 5.2.1 to 5.2.3 are set to zero: C_{nis}, $SALVA_n$, UC_{nsk}, TN_{nsk}, TG_{nsk}, UT_{nsk}, $UTOC_{nsk}$, AC_n, EXB_{yn}, and EXC_{yn}.

5.3 Economic decision criteria

5.3.1 Indicators determined

The following economic indicators are computed from the time streams of benefits or costs at the user-specified discount rate:

- **Net Present Value** - NPV (see Section 5.3.2)
- **Internal Rate of Return** - IRR (see Section 5.3.3)
- **Benefit/Cost Ratio** - BCR (see Section 5.3.4)
- **First Year Benefits** - FYB (see Section 5.3.5)

The determination of these indicators is described in the sections referenced.

5.3.2 Net present value

The Net Present Value (NPV) of investment option *m* relative to base option *n* is the sum of the discounted annual net benefits or costs, calculated from the relationship:

$$NPV_{(m-n)} = \sum_{y=1}^{Y} \frac{NB_{y(m\text{-}n)}}{[1+0.01*r]^{(y-1)}} \qquad \text{...(5.24)}$$

where:

$NB_{y(m\text{-}n)}$	net economic benefit of investment option *m* relative to base option *n* in year *y*
r	discount rate (%)
y	analysis year (y = 1, 2, ..., Y)

The higher the NPV, the greater the benefits from investment option *m* relative to base option *n*. If there are no budget constraints, then the choice between the two alternative investments should be based on NPV. Larger investments will tend to have larger NPVs.

5.3.3 Internal rate of return

The Internal Rate of Return (IRR) is the discount rate at which NPV is zero. It is calculated by solving the implicit relationship for $r°$:

$$\sum_{y=1}^{Y} \frac{NB_{y(m-n)}}{[1+0.01*r°]^{(y-1)}} = 0 \qquad ...(5.25)$$

This equation is solved for $r°$ by evaluating the NPV at 5 percent intervals of discount rates between -95 and +900 percent, and determining the zero(es) of the equation by linear interpolation of adjacent discount rates with NPV of opposite signs. Depending on the nature of the net benefit stream, $NB_{y(m-n)}$, it is possible to find one solution, multiple solutions, or none at all.

The IRR gives no indication of the size of the costs or benefits of an investment; it acts as a guide to the **profitability** of the investment - the higher the better. If the computed IRR is larger than the planning discount rate, then the investment is economically justified.

5.3.4 Benefit cost ratio

The Benefit Cost Ratio (BCR) of investment option *m*, relative to base option *n*, is the ratio is calculated as follows:

$$BCR_{(m-n)} = \frac{NPV_{(m-n)}}{C_m} + 1 \qquad ...(5.26)$$

where:

$BCR_{(m-n)}$	benefit cost ratio of investment option *m* relative to base option *n*
$NPV_{(m-n)}$	discounted total net benefit of investment option m relative to base option *n*. This is the Net Present Value at discount rate *r*
C_m	discounted total agency costs of implementing investment option *m*

If the $NPV_{(m-n)}$ is zero, then $(NPV/C)_{(m-n)}$ is zero. These ratios give an indication of the **profitability** of investment option *m* relative to base option *n* at a given discount rate. These measures eliminate the bias of NPV towards larger project options but, like the IRR, they give no indication of the size of the costs or benefits involved.

5.3.5 First-year benefits

The First-Year Benefits (FYB) is defined as the ratio, in percent, of the net benefit realised in the first year after construction (or improvement) completion to the increase in total capital cost:

$$FYB_{(m-n)} = \frac{100 * NB_{y°(m-n)}}{\Delta TCC_{(m-n)}} \qquad ...(5.27)$$

where:

$FYB_{(m-n)}$	first-year benefits of investment option *m* relative to base option *n* (%)
$NB_{y°(m-n)}$	net economic benefit of investment option *m* relative to base option *n* in year *y*°, where:
	y° is the year immediately after the last year in which the capital cost for improvement or construction is incurred in option *m*
$\Delta TCC_{(m-n)}$	the difference in total capital cost (non-discounted) of investment option *m* relative to base option *n*

FYB gives a rough guide to project timing: if it is greater than the discount rate, then the project should go ahead; otherwise it should be delayed until it satisfies the criterion.

5.4 Comparison of environmental effects

Where it is not possible to model costs directly, the **effects** of alternative investments can be evaluated. This information could be used as a decision tool for screening projects. For example; investment alternatives can be selected that are more effective in reducing the number of fatal accidents, or which are more effective in reducing the number of persons disturbed by a high level of traffic noise. Effects may also be a useful input for multi-criteria analysis.

The approach to carrying out a **comparative study** of environmental effects, for a pair of investment options, is similar to that used for economic analysis (see Section 5.2). The annual net quantities of vehicle emissions, number of accidents, and levels of traffic noise that are determined are compared with the benefit of implementing one investment option relative to the other.

5.5 Diverted traffic

Traffic diversion reduces or increases traffic on the roads that are affected. Therefore, in a situation where a road works causes traffic to divert significantly to a new or improved road section, a direct economic comparison of section options is not valid since the normal traffic flows on the road with and without the works are not identical.

Economic comparisons of investment options involving diverted traffic can only be performed meaningfully at the project analysis level, if the following conditions are met:

- All the road sections from and to which traffic diverts must be analysed together with the section(s) being considered under the investment analysis; this implies that a study area be defined to comprise all the sections that are affected significantly by traffic diversion as a result of carrying out the road works.
- In any given analysis year, the total traffic volume entering the study area equals the total traffic volume exiting the area; this implies a fixed trip matrix.

The analysis of a new road section (or link) in an entirely new location always involves diverted traffic. The **normal** traffic in the first year of road opening is diverted traffic from nearby routes (and from other transport modes, which may complicate matters further). The economic analysis and comparison involving a new road section, therefore, should always comply with the conditions described above.

6 Optimisation

The two methods for budget optimisation provided for road works programming and network strategic analysis are:

1 **Total enumeration**

2 **Incremental benefit/cost ranking**

[Further methods may be added in later versions of the HDM-4 software.]

If the number of roads to be analysed is less than 100, and there are no more than five budget periods and 16 alternatives per road, total enumeration (see Section 6.1) can be used. This will be externally done in the EBM-HS model of HDM-III. If the above mentioned constraints are exceeded, incremental benefit/cost ranking (see Section 6.2) will be used.

6.1 Total enumeration

This is the method used by the EBM-HS model of HDM-III. It requires the user to specify the following parameters:

- **Name of data set**

  For example, **CAPROG97**.

- **Length of analysis period**

  For example, 20 years.

- **Budget periods**

  For example, 1, 2, 3, 4-20 years.

- **Objective function**

  Either: maximise NPV or maximise the improvement in roughness.

- **Constraints on resources for each budget period**

  For example, 10, 10, 10, 200.

The analysis period is given in terms of the number of years over which the overall analysis should be performed, together with the initial calendar year. Budget periods are shorter time periods for which the budget constraints are given. The objective function defines which parameter is to be optimised. The default is the maximisation of NPV over the analysis period, but the user can also choose the maximisation of the improvement in roughness.

The program is run for all the road sections defined with positive economic return, and for all budget and investment options. The budget requirements from any committed projects will be deducted from the available budget and the balance is used for optimisation.

The optimisation problem is then defined as an integer programming problem of maximising the total objective function (TOBJ) for the network (*extracts from EBM documentation*):

$$\text{Maximise } TOBJ[X_{sm}] = \sum_{s=1}^{S} \sum_{m=1}^{M_s} OBJ_{sm} X_{sm} \qquad \text{...(6.1)}$$

where:

s a road section (s = 1, 2, ... , S)

M_s the number of alternatives for road section *s*

m an investment alternative on a road section

OBJ_{sm} the objective function to be maximised which may be the discounted net present value of economic benefits, or the average reduction in roughness due to the investment alternative

sm subscript denoting alternative *m* for road section *s*

X_{sm} the **zero-one** decision variable:

X_{sm} 1 if alternative *m* of investment unit *s* is chosen

X_{sm} 0 otherwise

m 1,...,M_s

The above is subject to the following resource constraints:

$$\sum_{s=1}^{S}\sum_{m=1}^{M_s} R_{smqt}\ X_{sm} \le TR_{qt},\ q = 1,....,\ Q;\ \ t = 1,....,\ T \qquad ...(6.2)$$

where:

R_{smqt} Non-discounted amount of resource of type *q* incurred by the sectoral agency within a budget period *t*

TR_{qt} maximum amount of resource type *q* available for budget period *t*

Q the total number of resource types

T the total number of budget periods (the duration of *t* may be one or more years and need not be equal for different budget periods)

The above is subject to the constraint of mutual exclusivity:

$$\sum_{m=1}^{M_s} X_{sm} \le 1,\ s = 1,....,\ S \qquad ...(6.3)$$

that is, for each road section *s*, no more than one alternative can be implemented.

If *M* is the average number of alternatives for the roads, the problem then has SM (= S x M) zero-one variables, *QT* (= Q x T) resource constraints and *S* interdependency constraints. The parameters that define the problem size are *S*, *M* and *QT*. Depending on the solution method used, different problem-size parameters determine whether the method is suitable for the problem in terms of the computational effort needed.

The **total enumeration** method provides the user with an unconditionally optimal solution. It computes the total net present values of all feasible programme selections, and chooses the one with the highest value. The computational effort required for this may be considerable, so the method is only feasible when the number of alternatives per investment unit is relatively small.

Total enumeration is performed externally in the EBM-HS software of HDM-III. The procedure is as follows:

1 Create an input file for the EBM-HS in the programme analysis. The format of this file is defined in the HDM-III EBM documentation.

2 The user starts the EBM-HS, imports the file to the EBM-HS, runs it, and exports results to an output file.

3 The output file is imported to the programme analysis for reporting.

6.2 Incremental benefit/cost ranking

With many applications of HDM-4, a large number of road sections will need to be prioritised. In these cases, the incremental benefit/cost method is the most appropriate. This involves searching through investment options on the basis of the incremental NPV/cost ratio of one alternative compared against another. The incremental NPV/cost is defined as:

$$E_{ji} = \left[\frac{\left(NPV_j - NPV_i \right)}{\left(cost_j - cost_i \right)} \right] \qquad \text{...(6.4)}$$

where:

E_{ji}	the incremental NPV/cost ratio
NPV_j	the net present value of the more expensive alternative *j*
NPV_i	the net present value of the cheaper alternative *i*
$cost_j$	the economic cost of the more expensive alternative *j*
$cost_i$	the economic cost of the cheaper alternative *i*

In Equation 6.4 above, the incremental NPV/cost can be replaced by the incremental ΔIRI*Length/cost where ΔIRI*Length is the weighted average change in roughness obtained by comparing the project alternatives using IRI instead of NPV.

The objective of the incremental method is to select road sections successively starting with the largest NPV/cost ratio (E_{ji}), since this maximises the NPV (net present value) for any given budget constraint. Where there is more than one investment option on any individual road section, that with the lowest discounted investment costs is designated the **base case** alternative. This method considers all possible options, and compares these incrementally starting against the base case, by using the incremental algorithm to select the combination that maximises the selected objective function.

An incremental search technique is used to select the options with successively lower incremental NPV/cost ratios, ensuring that at any time there is no more than one option per road section. The process continues until the budget is exhausted for each budget period. The method is often referred to as the **efficiency frontier**, which is a **line** that joins investments with the highest NPV along the cost axis in a plot of NPV against investment cost (*Harral and Faiz, 1979*). In essence, the method seeks out those options that are close to the boundary of the efficiency frontier. The algorithm is illustrated in Figure G1.3, and is defined in the following steps:

1. Determine the pre-defined investment options for pre-selected sections and deduct the **financial** costs of these options from the available budget in corresponding years. Exclude these sections from any further optimisation.

2. Determine possible investment options for the remaining sections. If the life cycle analysis option is being used, set the user-defined base alternatives as the **do minimum** for each road section. For the multi-year forward programme, the **do minimum** option is that with the delayed capital works.

3. If the total **financial** cost of the **do minimum** investment alternatives on each section is greater than the available budget for any period, then the investment options or budget constraints must be redefined.

4. Deduct the **financial** cost of the **do minimum** investments from the available budget to determine the remaining budget for each period. Set the **do minimum** as the first **Base option** for each section.

5. Calculate the incremental NPV/Cost ratio for all remaining section-options compared against the Base option, and all other option pairs with higher **economic** cost. For example, consider the following investment options for a particular section arranged in the ascending order of discounted total **economic** costs:

   options: A, B, C, D, E

   The incremental NPV/Cost ratios for these are given by:

   Eba Eca Eda Eea ; Ecb Edb Eeb ; Edc Eec ; Eed

6 Delete incremental NPV/cost ratios that are less than the user specified minimum incremental value (MIV).

7 List the remaining incremental NPV/cost ratios in decreasing order (with the associated section-option pair codes) and, within each incremental NPV/cost, in the order of decreasing **economic** cost. For example, if $E_{eb} = E_{db}$ then E_{eb} is ranked higher.

8 Select the next incremental NPV/cost ratio from the top of the list. If the lower cost section-option is not the current Base Option for that section, continue searching until one is found.

9 If the remaining budget is insufficient in any of the periods for the **financial** costs of works required for the section-option selected in Step 8 above, then the selected option should be rejected, and continue searching by repeating Step 8.

10 If the section-option can fit within the remaining budgets for all periods, deduct the net **increase in financial** cost of capital works from all corresponding budget periods. Set the Base option for this section to be that corresponding to the lower cost option for the incremental NPV/Cost ratio chosen in Step 8. Providing that the remaining list is not empty, return to Step 8.

The process described above continues until the budget is exhausted or there are no more section-options remaining in the list. The resulting list of selected section-alternatives constitutes the optimal work programme.

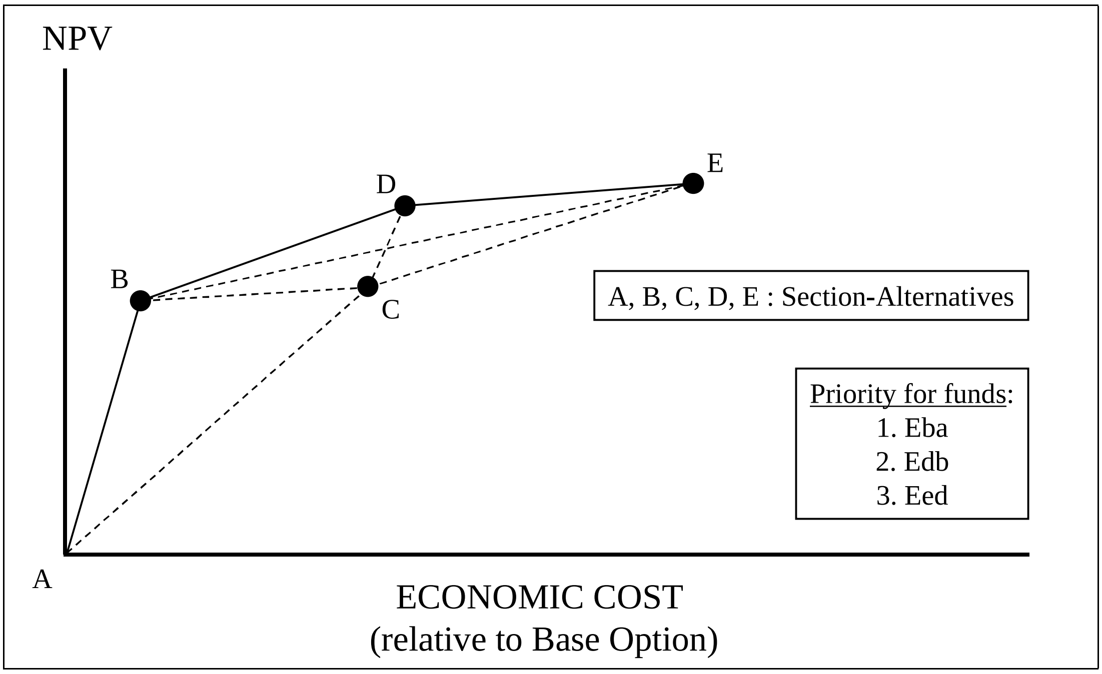

Figure G1.3 Efficiency frontier concept

7 References

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